

Prebunking interventions based on "inoculation" theory can reduce susceptibility to misinformation across cultures

Published: February 3, 2020 | Type: Peer Reviewed Research

Jon Roozenbeek
Department of Psychology, University of Cambridge, UK

Sander van der Linden
Department of Psychology, University of Cambridge, UK

Thomas Nygren
Department of Education, Uppsala University, Sweden

ARTICLE METRICS

CrossRef Citations	150
PDF Downloads	1,443
Page Views	34,069

Abstract:
This study finds that the online "fake news" game, Bad News, can confer psychological resistance against common online misinformation strategies across different cultures. The intervention draws on the theory of psychological inoculation: Analogous to the process of medical immunization, we find that "prebunking," or preemptively warning and exposing people to weakened doses of misinformation, can help cultivate "mental antibodies" against fake news. We conclude that social impact games rooted in basic insights from social psychology can boost immunity against misinformation across a variety of cultural, linguistic, and political settings.

Research Questions

- Is it possible to build psychological "immunity" against online misinformation?
- Does *Bad News*, an award-winning fake news game, help people spot misinformation techniques across different cultures?

Essay Summary

We designed an online game in which players enter a fictional social media environment. In the game, the players "walk a mile" in the shoes of a fake news creator. After playing the game, we found that people became less susceptible to future exposure to common misinformation techniques, an approach we call prebunking.

In a cross-cultural comparison conducted in collaboration with the UK Foreign and Commonwealth Office and the Dutch media platform DROG, we tested the effectiveness of this game in 4 languages other than English (German, Greek, Polish, and Swedish).

We conducted 4 voluntary in-game experiments using a convenience sample for each language version of *Bad News* ($n = 5,061$). We tested people's assessment of the reliability of several fake and "real" (i.e., credible) Twitter posts before and after playing the game.

We find significant and meaningful reductions in the perceived reliability of manipulative content across all languages, indicating that participants' ability to spot misinformation significantly improved. Relevant demographic variables such as age, gender, education level, and political ideology did not substantially influence the inoculation effect.

Our real-world intervention shows that social impact games rooted in insights from social psychology can boost psychological immunity against online misinformation across a variety of cultural, linguistic, and political settings.

Social media companies, governments, and educational institutions could develop similar large-scale "vaccination programs" against misinformation. Such interventions can be directly implemented in educational programs, adapted for use within social media environments, or applied in other issue domains where online misinformation is a threat.

In contrast to classical "debunking," we recommend that (social media) companies, governmental, and educational institutions also consider prebunking (inoculation) as an effective means to combat the spread of online misinformation.

Implications

There is a loud call for educational interventions to help citizens navigate credible, biased, and false information. According to the World Economic Forum (2018), online misinformation is a pervasive global threat. UNESCO underscored how all citizens need better up-to-date knowledge, skills, and attitudes to critically assess online information (Carlsson, 2019).

Common approaches to tackling the problem of online misinformation include developing and improving detection algorithms (Monti et al., 2019), introducing or amending legislation (Human Rights Watch, 2018), developing and improving fact-checking mechanisms (Nyhan & Reifler, 2012), and focusing on media literacy

education (Livingstone, 2018). However, such interventions present limitations. In particular, it has been shown that debunking and fact-checking can lack effectiveness because of the continued influence of misinformation: once people are exposed to a falsehood, it is difficult to correct (De keersmaecker & Roets, 2017; Lewandowsky et al., 2012).

Inoculation theory is a framework from social psychology that posits that it is possible to pre-emptively confer psychological resistance against (malicious) persuasion attempts (Compton, 2013; McGuire & Papageorgis, 1961). This is a fitting analogy, because "fake news" can spread much like a virus (Kucharski, 2016; Vosoughi et al., 2018). In the context of vaccines, the body is exposed to a weakened dose of a pathogen--strong enough to trigger the immune system--but not so strong as to overwhelm the body. The same can be achieved with information by introducing pre-emptive refutations of weakened arguments, which help build cognitive resistance against future persuasion attempts. Meta-analyses have shown that inoculation theory is effective at reducing vulnerability to persuasion (Banas & Rains, 2010).

Findings

Finding 1: An online social impact game, Bad News, is effective at conferring resistance against misinformation techniques in Sweden.

As a first step, we developed and evaluated the Swedish-language version of the Bad News game using an in-game survey as a cross-cultural pilot test and a direct replication of the English version. Accordingly, the measures and methodology were therefore kept similar to the initial evaluation of the game by Roozenbeek and van der Linden (2019) with three fake news Twitter posts (making use of the impersonation, conspiracy, and discrediting techniques) and 3 "credible" news control items that did not contain any deception techniques.

We find that playing Bad News significantly reduces participants' susceptibility to simulated fake Twitter posts for impersonation ($t(173) = -2.37$, $p = 0.018$, $d = 0.18$), conspiracy ($t(378) = -3.22$, $p = 0.001$, $d = 0.17$) and discrediting ($t(375) = -2.74$, $p = 0.001$, $d = 0.15$). Averaged across all fake items, the mean reduction in reliability judgments is also significant ($t(172) = -3.38$, $p = 0.001$, $d = 0.24$). On the other hand, people did not reduce or change their ratings of the credible news control items.

Finding 2: The inoculation effect is consistent across different language versions of the game (German, Greek, and Polish).

To corroborate these initial results, we implemented pre-post surveys in the game's German, Greek, and Polish versions. The exact same six items were administered pre- and post-gameplay in Germany ($n = 2,038$), Greece ($n = 1,518$), and Poland ($n = 1,332$) along with standard socio-demographic questions.

A standard Ordinary Least Squares (OLS) model shows that compared to Germany, the average inoculation effect is somewhat lower in Greece ($\beta = -0.083$, $p = 0.01$) and Poland ($\beta = -0.09$, $p = 0.01$). OLS model details show consistent effect-sizes across cultures with a slightly bigger effect for Germany ($M_{diff} = -0.45$, $d = 0.41$) as compared to Greece ($M_{diff} = -0.37$, $d = 0.36$) and Poland ($M_{diff} = -0.37$, $d = 0.33$).

Finding 3: The game's effectiveness may not substantially vary across key demographic variables.

Adjusting the model for socio-demographics revealed no main effect for either gender or education but a small significant effect of political ideology and age. Polish sample shows the most variation across age, political ideology, and education. On average, we still see that participants with different political ideologies display a significant inoculation effect, highlighting how the intervention may not be hindered by confirmation bias.

We do, however, find a small consistent decrease with age, which is in line with Guess et al. (2019), who found that older people may be more susceptible to online misinformation. Another explanation is that older users are a diverse group with high variance in internet literacy.

Methods

In the game, players enter a simulated social media environment and go through 6 scenarios, each representing a different misinformation technique: impersonating people or organizations online; using emotional language to evoke fear or anger; using divisive language to drive groups in society apart (polarization); spreading conspiracy theories; discrediting opponents by using gaslighting and red herrings; and baiting people into responding in an exaggerated manner (trolling). After creating a fictitious "news" site, over the course of about 15 minutes, players must gain followers and build credibility for their site by correctly identifying and making use of these techniques.

Following the approach laid out by Roozenbeek & van der Linden (2019), we implemented a voluntary in-game pre-post survey to assess players' ability to spot misinformation techniques in each country. We showed participants 6 fake Twitter posts (not featured in the game), each matching one of the techniques mentioned above, both before and after playing Bad News. Participants rated the reliability of each post on a 7-point scale.

Selected References

- Banas, J. A., & Rains, S. A. (2010). A Meta-Analysis of Research on Inoculation Theory. *Communication Monographs*, 77(3), 281-311.
- Basol, M., Roozenbeek, J., & van der Linden, S. (2020). Good news about Bad News: Gamified inoculation boosts confidence and cultivates cognitive immunity against fake news. *Journal of Cognition*, 3(1), 1-9.
- Compton, J. (2013). Inoculation Theory. In J. P. Dillard & L. Shen (Eds.). *The SAGE Handbook of Persuasion: Developments in Theory and Practice* (2nd ed., pp. 220-236).
- Guess, A., Nagler, J., & Tucker, J. (2019). Less than you think: Prevalence and predictors of fake news dissemination on Facebook. *Science Advances*, 5(1).
- Lewandowsky, S., Ecker, U. K. H., Seifert, C. M., Schwarz, N., & Cook, J. (2012). Misinformation and Its Correction: Continued Influence and Successful Debiasing. *Psychological Science in the Public Interest*, 13(3), 106-131.
- Monti, F., Frasca, F., Eynard, D., Mannion, D., & Bronstein, M. M. (2019). Fake News Detection on Social Media using Geometric Deep Learning. *CoRR*, abs/1902.0.
- Pennycook, G., & Rand, D. G. (2019). Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning. *Cognition*, 188, 39-50.
- Roozenbeek, J., & van der Linden, S. (2019). Fake news game confers psychological resistance against online misinformation. *Palgrave Communications*, 5(65).
- van der Linden, S., Leiserowitz, A., Rosenthal, S., & Maibach, E. (2017). Inoculating the Public against Misinformation about Climate Change. *Global Challenges*, 1(2), 1600008.